

Alexander Joos

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EDUCATION

Cornell University

MEng Computer Science, GPA: 3.65

Ithaca, NY

Jan. 2026 – Dec. 2026

Cornell University

B.A. Information Science (Data Science)

Ithaca, NY

Aug. 2022 – Dec. 2025

National University of Singapore

Yale-NUS College Exchange Program (Computer Science)

Singapore

Jan. 2025 – May 2025

Graduate Coursework: Algorithms, Systems Programming in Rust, Distributed ML Systems (LLM Training & Inference), Software Engineering, Machine Learning, Deep Learning, LLM Applications (RAG & AI Agents), Market Microstructure

SKILLS

Languages: Python, Java, C/C++, SQL, Rust, OCaml, JavaScript, TLA+

ML & AI: PyTorch, Hugging Face, scikit-learn, LangChain, LLM APIs, MCP, RAG, Vector DBs, Multi-Agent Systems

Data: Kafka, Aeron, Airflow, pandas, SingleStore, Snowflake, BigQuery, PostgreSQL, Oracle, CockroachDB

Backend & Cloud: Flask, FastAPI, REST APIs, SQLAlchemy, Docker, AWS (S3), GCP, Linux, Git, Bash, CI/CD, Jira

EXPERIENCE

Fidelity Investments

June 2026 – Aug. 2026

Graduate Data Engineering Intern, Database Modernization

Merrimack, NH

- Built Python migration agents moving COBOL-era APIs & legacy data into an event-sourced ledger via a four-pass design.
- Developed validation agents & tool-call workflows checking migrated output with TLA+/TLC specs & Python checks.
- Wrote high-throughput Java Kafka producer over Aeron with async fault-tolerant delivery for a sub-3ms brokerage platform.
- Built Aeron/Kafka pipelines streaming a deterministic brokerage platform's state machine output into SingleStore.

Fidelity Investments

June 2025 – Aug. 2025

Data Engineering Intern, Cloud and Platform Engineering

Merrimack, NH

- Owned & launched a Flask/AJAX web app on an Ubuntu VM to automate internal database model comparison.
- Designed Flask REST API endpoints & Python/SQL backend storage/retrieval behind a JavaScript/HTML/CSS frontend.
- Deployed the application through Apache, SSL certificate setup, firewall whitelisting, & DEV/PROD rollout.
- Supported brokerage platform model analysis across Oracle, PostgreSQL, CockroachDB, & Snowflake databases.

American Family Insurance: Homesite

June 2023 – Aug. 2024

Partner Distribution Intern → Data Analytics Intern

Boston, MA

- Built BigQuery SQL pipelines from raw source tables for a Fortune 50 retailer's Partner Management reporting layer.
- Built 8 Looker dashboards for Partner & Executive teams on top of the BigQuery reporting layer.
- Automated SQL query generation with Excel/Sheets scripts, replacing manual query writing for recurring reports.

PROJECTS

BME Collaborator Finder – Cornell AI Innovation Lab

Sep. 2026 – Present

- Building multi-agent collaboration system where researcher-persona agents converse to surface Cornell research partnerships.
- Developing agentic RAG foundation: PDF & PubMed ingestion into S3, vector search, & cited faculty matching.

Apache Airflow Anomaly Detection – Agile Open Source Systems & Backend Feature Development

May 2026

- Built Airflow-native task-runtime anomaly detection with DAG-file config, custom detectors, & 4 built-in algorithms.
- Implemented backend/task SDK, structured results, task-level observability, CI tests, & example DAGs/docs.

LOB-Driven Optimal Execution Strategy – Quantitative Trading & Market Microstructure

May 2026

- Beat TWAP on INTC/MSFT/AMZN/GOOG & held-out AAPL using order-book imbalance, spread, & timing signals.
- Generalized across names by fitting parameters per spread regime via smoothed grid search on tick-level LOB data.

LLM Text Watermarking – ML Research Reimplementation

May 2026

- Reimplemented Kirchenbauer et al. LLM watermarking in PyTorch with soft green-list logit biasing & z-test detection.

Coursly – Full-Stack Course Review App

Nov. 2025

- Built backend for a Cornell course review app: Flask REST APIs, scraper, SQLite model, Docker Compose, GCP deploy.